Part D: GitHub Repository Initialization

Universe 25 Population Simulator, APPDAET Final Project, Group 2

GitHub Link: https://github.com/MAML623/APPDAET-Group2-Universe25-Population-Simulator.git

D.1 Create and Configure the GitHub Repository

The group created a public GitHub repository for the Universe 25 Population Simulator. The repository is used to store the Python project files, document progress, and allow members to work on the same application. It uses the main branch and includes the initial README.md and Python .gitignore files.

Repository Information	Details
Repository name	APPDAET-Group2-Universe25-Population-Simulator
Repository owner	MAML623
Visibility	Public
Default branch	main

Repository URL: https://github.com/MAML623/APPDAET-Group2-Universe25-Population-Simulator

D.2 README.md and .gitignore

The README.md explains the project overview, planned application features, simulation algorithm, technologies, proposed project structure, input and output requirements, limitations, group members, and AI assistance disclosure. The Python .gitignore file helps prevent temporary Python files, virtual environments, and other unnecessary development files from being uploaded to the repository.

D.3 Initial Project Folder Structure

The group organized the repository based on the planned application structure in Part C. The Python files currently contain descriptions of their intended responsibilities; the actual simulation functions will be developed in the next stages.

File / Folder	Proposed Responsibility
main.py	Starts the Python desktop application.
model.py	Calculates births, deaths, and changes in resources and population.
validation.py	Checks user inputs and rejects invalid settings.
state.py	Keeps the simulation state and manages Run, Step, and Reset.
gui.py	Handles the Tkinter window, generation table, and Matplotlib graphs.
tests/	Stores testing files and automated tests.
docs/	Stores project planning documents and reports.

D.4 Upload Initial Planning Documents

The group uploaded the available planning documents inside the docs/ folder: Part A - Project Understanding and Scope, Part B - Algorithm Design and Program Decomposition, Part C - Development Environment and Technical Preparation, and Part E - Project Management.

D.5 Demonstrate the Basic Git Workflow

The group demonstrated the four required Git operations using GitHub and Visual Studio Code on macOS. The following actions were performed:

Operation	Action Performed
Clone	Copied the GitHub repository to the Mac and opened it in VS Code.
Commit	Edited main.py, staged the change, and saved it as a local commit.
Push	Uploaded the local main.py commit to the GitHub repository.
Pull	Updated README.md on GitHub and downloaded the change into the local VS Code copy.

The commit and synchronization history in VS Code confirmed that the local and online copies were updated successfully. These steps show how the group can use Git to manage future changes to its Python code.

D.6 Add Group Members and Grant Access

The repository owner invited the group members to collaborate, and the invited group members have accepted their invitations. This enables the members to contribute changes to the same project repository. The repository remains public, but only users with the appropriate permissions can directly modify the files.

D.7 Evidence Documentation

Screenshots of the repository setup, initial folders and planning files, and the Clone, Commit, Push, and Pull operations are attached after this section as Figures 1-7. A separate screenshot of collaborator permissions should be added when available.

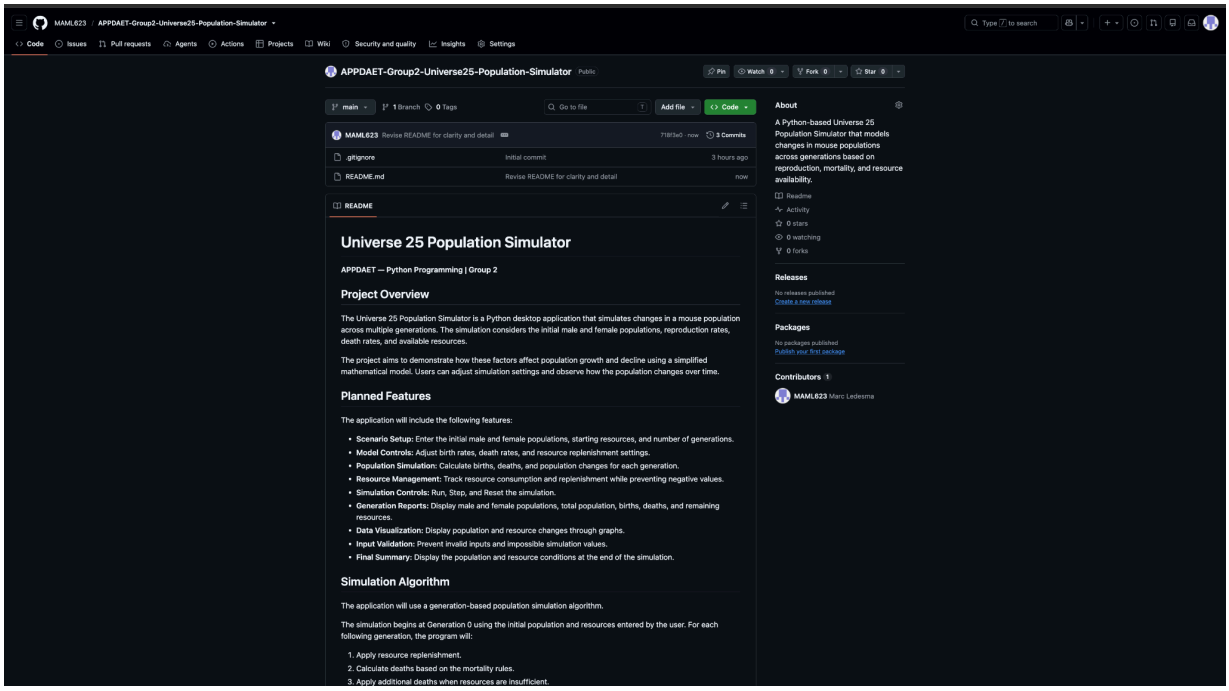
D.8 AI Assistance Disclosure

ChatGPT was used to help organize the initial GitHub repository, prepare README descriptions, explain the Git workflow, and compile this documentation. The group performed the GitHub and VS Code steps and remains responsible for its repository, files, and Python implementation.

Appendix: GitHub Initialization Evidence

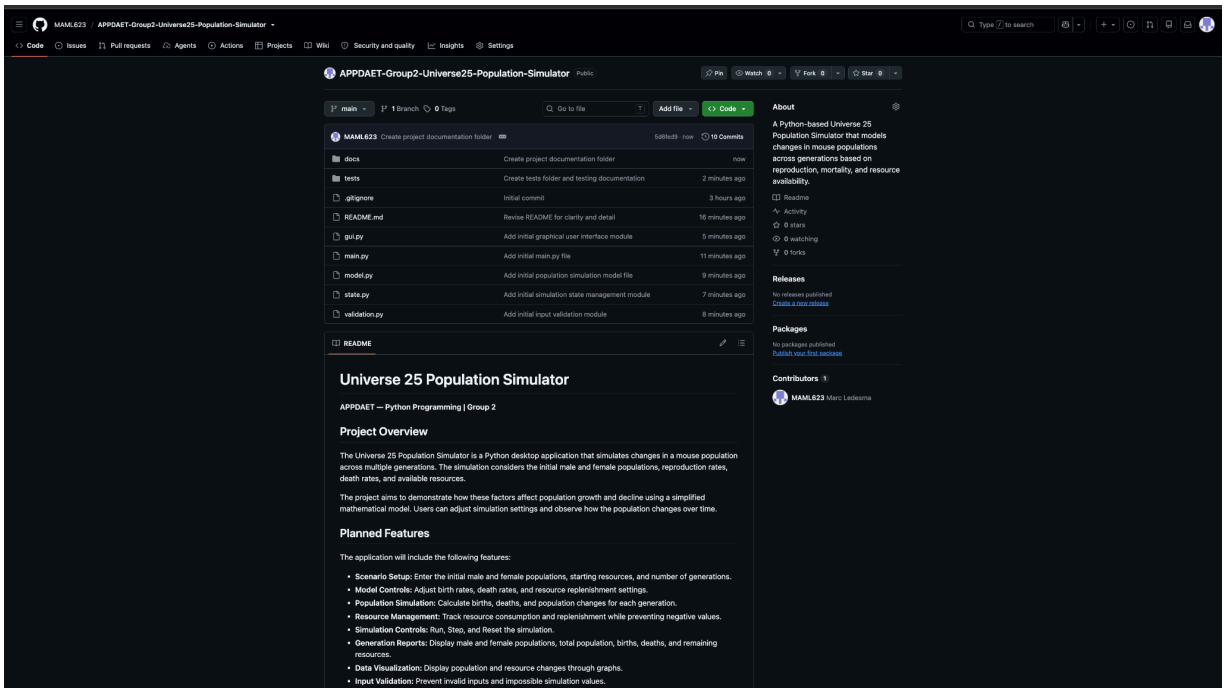
The following screenshots were captured during the repository setup and Git workflow demonstration.

Figure 1. GitHub repository initialization



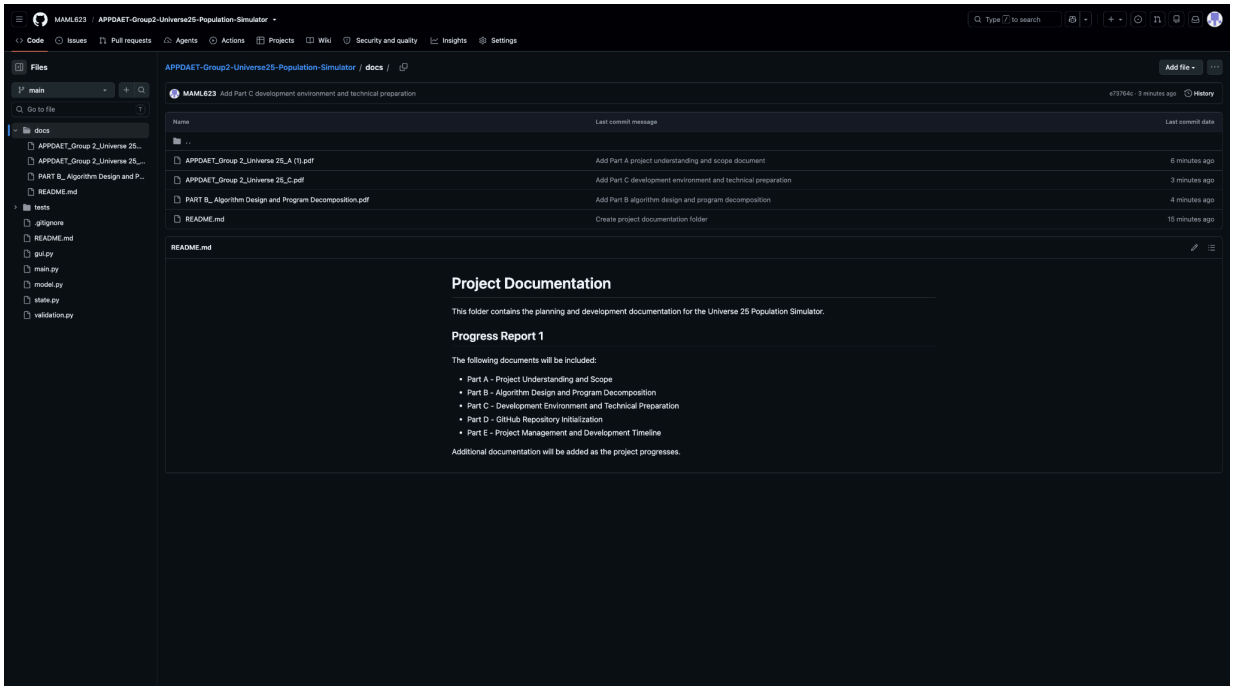
Repository home page showing the public repository, README.md and .gitignore.

Figure 2. Initial project structure



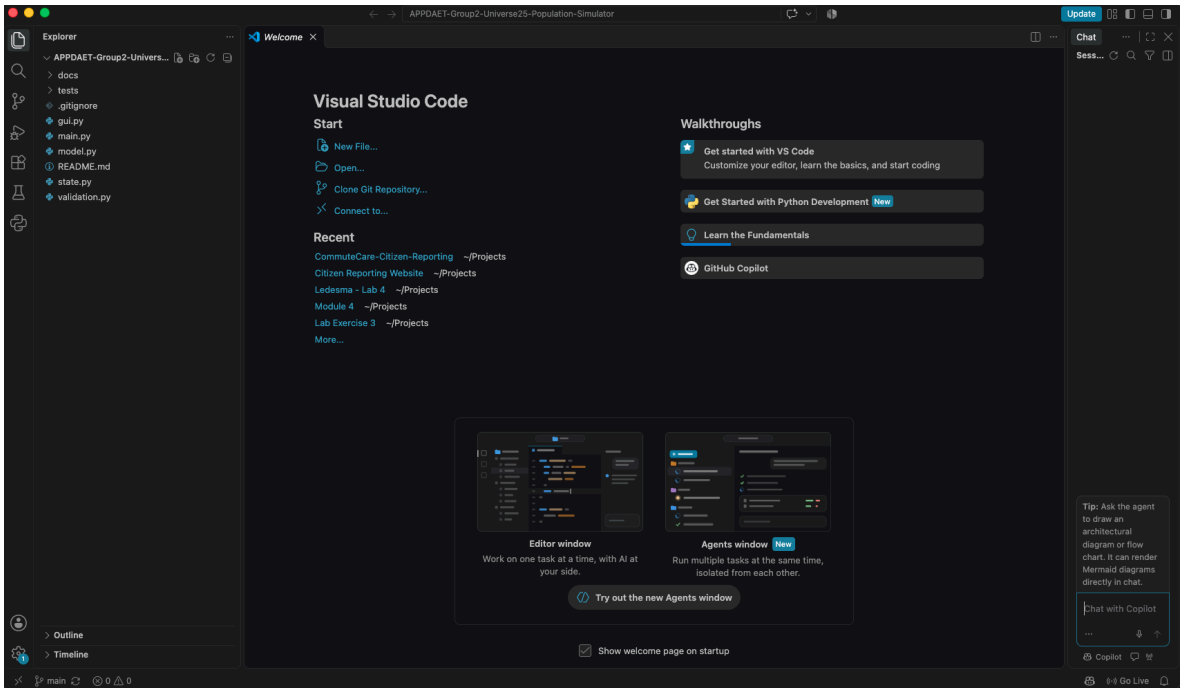
Project folders and the five Python module files.

Figure 3. Uploaded planning documentation



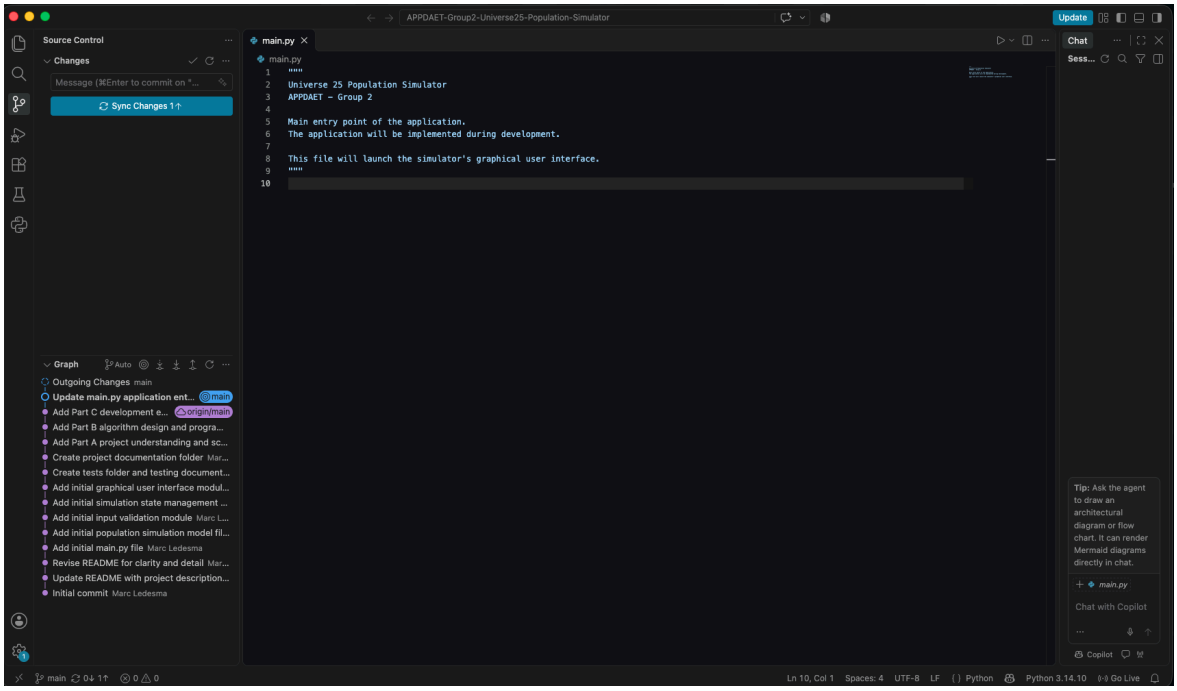
docs/ folder containing the Parts A, B and C PDF documents.

Figure 4. Repository cloning



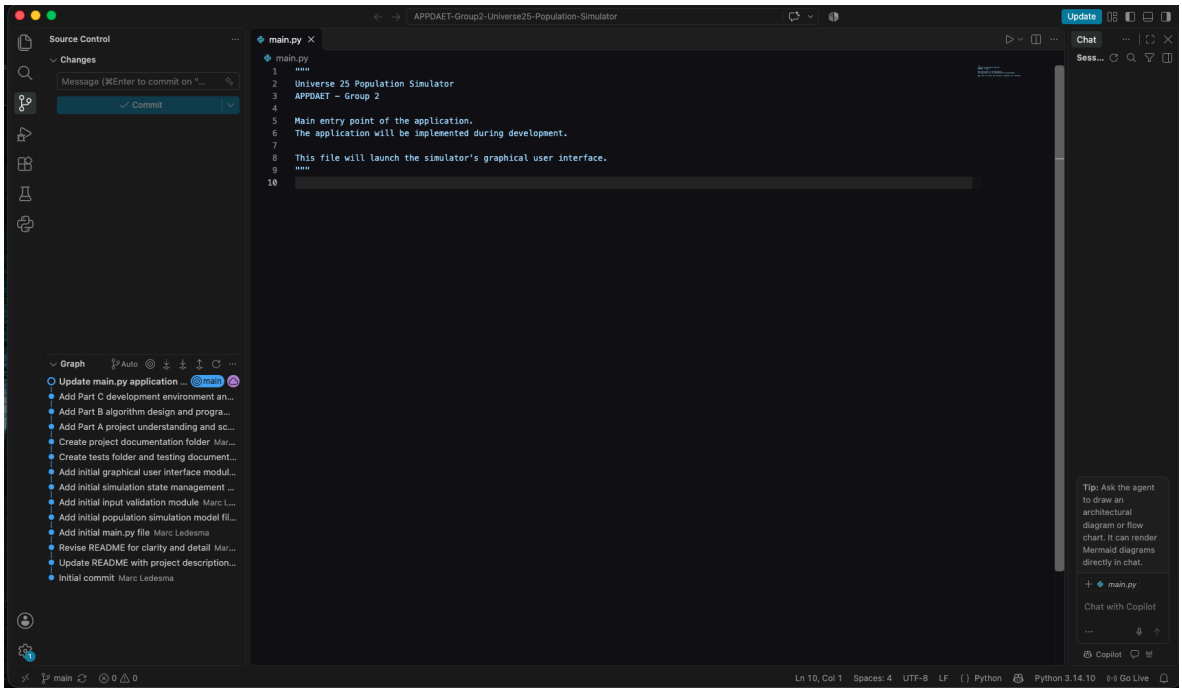
GitHub repository cloned and opened in Visual Studio Code.

Figure 5. Local Git commit



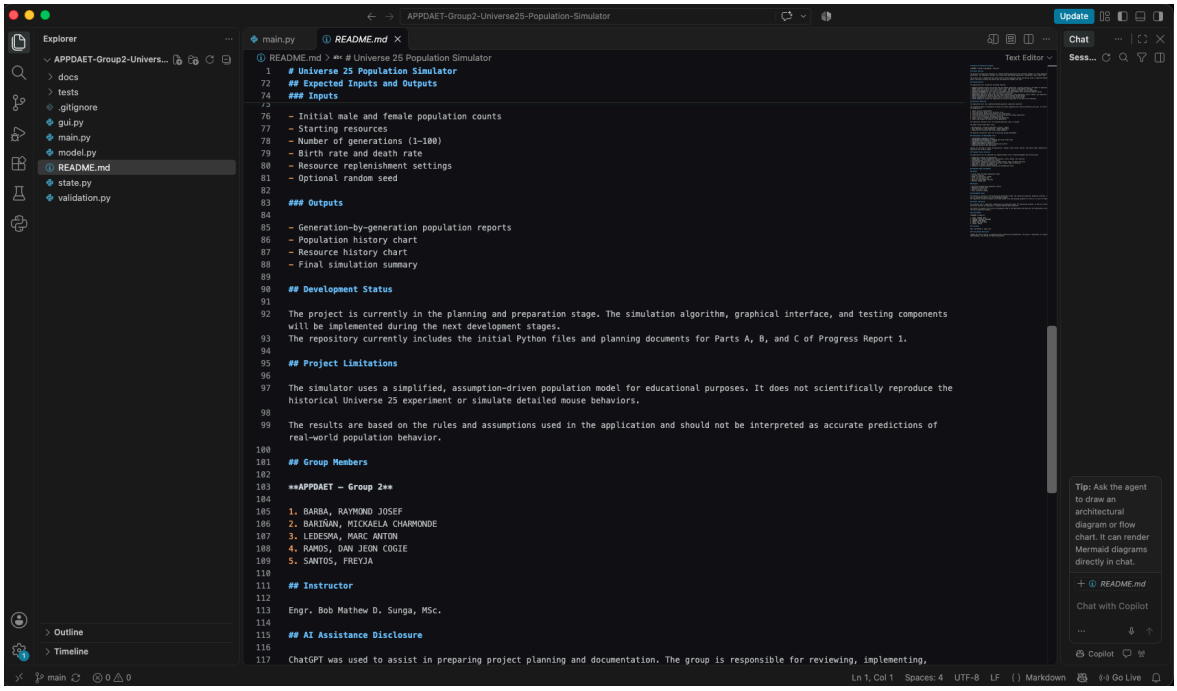
Source Control showing the locally committed main.py change and outgoing commit.

Figure 6. Git push



Source Control after pushing; the outgoing change indicator is no longer present.

Figure 7. Git pull



README.md in VS Code showing the new sentence committed through GitHub.